Tyler Sheaves

sheaves94@gmail.com | (415) 279-2195 | San Jose, CA | Google Scholar | LinkedIn

Summary

Ph.D. candidate specializing in hardware security, side-channel analysis, and secure reconfigurable computing. Experienced in FPGA and embedded system design, digital design, and experimental hardware security research. Former intern with Intel's Programmable Solutions Group (now Altera).

Education

University of California, Santa Cruz — 2025–Present
Ph.D. in Computer Science and Engineering — *Defending June 2026*
Advisor: Prof. Dustin Richmond | Transferred from UC Davis

University of California, Davis — 2020–2025
Ph.D. program in Electrical and Computer Engineering

San Francisco State University — 2012–2017
B.S. in Electrical Engineering

Experience

Graduate Student Researcher, UC Santa Cruz — 2025–Present

- ◇ Discovered novel ReRAM timing side-channel on commercial crossbar memories.
- ◇ Designed SAT-assisted ECC reverse engineering technique using timing measurements.
- ◇ Demonstrated successful recovery of full ECC on two commercial ReRAM products.
- ◇ Evaluated commercial ReRAM as NIST-compliant entropy source.
- ◇ Designed conditioning and monitoring techniques for ReRAM-based TRNGs.

Graduate Student Researcher, UC Davis — 2020–2024

- ◇ Improved tuning resolution of time-to-digital converter side-channel sensor by 50x.
- ◇ Developed post-synthesis obfuscation techniques, taped out test chip in TSMC 65nm.
- ◇ Modeled novel hardware Trojan threat vectors in commercial IP supply chains.
- ◇ Developed obfuscation IP blocks using emerging memories (STT, ReRAM).

Graduate Technical Intern, Intel PSG (Altera) — 2020–2023

- ◇ Presented confidential computing infrastructure at the Intel Federal Summit.
- ◇ Integrated IPs with Open FPGA Stack on D5005 and N6000 PAC platforms.
- ◇ Developed FPGA GUI application used for academic and corporate trainings.

Lecturer & Teaching Assistant, UCSC / UCD / SFSU — 2017–2025

- ◇ TA for Digital I/II, Computer Architecture, Embedded Systems, and HW Security.
- ◇ Lecturer for Intro to MCUs and Digital Design Verification at SF State.

Technical Skills

HW Platforms: FPGA/SoC-FPGA (Altera), Embedded ARM & RISC-V, PAC D5005 & N6000.
Languages: Verilog & System Verilog for RTL & verification, Python, C/C++, Tcl/Tk.
Tools: Quartus/ModelSim, Cocotb, Yocto, SAT/SMT solvers, Synopsys digital IC suite, HSPICE.

Leadership & Service

Volunteer, Non-volatile Memories Workshop (NVMW) — 2026
Session Moderator, Workshop on Open-Source EDA Technology (WOSET) — 2024
Co-chair, Tutorials & Workshops, International Symposium on FPGAs (ISFPGA) — 2023

Publications

◇ **T. Sheaves**, A. Althoff, and D. Richmond, "BREW-RC: Bit-exact recovery of ECCs from write timing in ReRAM crossbars," in *2026 63rd ACM/IEEE Design Automation Conference (DAC)*, Accepted, IEEE, 2026.

◇ C. Drewes, **T. Sheaves**, O. Weng, K. Ryan, B. Hunter, C. McCarty, R. Kastner, and D. Richmond, "Turn on, tune in, listen up: Maximizing side-channel recovery in cross-platform time-to-digital converters," *ACM Transactions on Reconfigurable Technology and Systems*, 2024, Co-first author. DOI: 10.1145/3666092.

◇ A. Zeraatkar, P. Kamran, I. Kaur, N. Ramu, **T. Sheaves**, and H. Al-Asaad, "On the performance of malware detection classifiers using hardware performance counters," in *2024 International Conference on Smart Applications, Communications and Networking (SmartNets)*, IEEE, 2024, pp. 1–6. DOI: 10.1109/SmartNets61466.2024.10577644.

◇ C. Fang, N. Miao, H. Wang, J. Zhou, **T. Sheaves**, J. M. Emmert, A. Sasan, and H. Homayoun, "Gotcha! i know what you are doing on the fpga cloud: Fingerprinting co-located cloud fpga accelerators via measuring communication links," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2023, pp. 2024–2037. DOI: 10.1145/3576915.3616606.

◇ K. I. Gubbi, B. S. Latibari, A. Srikanth, **T. Sheaves**, S. A. Beheshti-Shirazi, S. M. Pd, S. Rafatirad, A. Sasan, H. Homayoun, and S. Salehi, "Hardware trojan detection using machine learning: A tutorial," *ACM Transactions on Embedded Computing Systems*, 2023. DOI: 10.1145/3579823.

◇ K. I. Gubbi, S. A. Beheshti-Shirazi, **T. Sheaves**, S. Salehi, P. D. S. Manoj, S. Rafatirad, A. Sasan, and H. Homayoun, "Survey of machine learning for electronic design automation," in *Great Lakes Symposium on VLSI*, 2022. DOI: 10.1145/3526241.3530834.

◇ G. Kolhe, **T. Sheaves**, D. S. M. P., H. Mahmoodi, S. Rafatirad, A. Sasan, and H. Homayoun, "Breaking the design and security trade-off of look-up-table–based obfuscation," *ACM Transactions on Design Automation of Electronic Systems*, vol. 27, 2022. DOI: 10.1145/3510421.

◇ G. Kolhe, **T. Sheaves**, K. I. Gubbi, S. Salehi, S. Rafatirad, P. D. S. Manoj, A. Sasan, and H. Homayoun, "Lock&roll: Deep-learning power side-channel attack mitigation using emerging reconfigurable devices and logic locking," in *59th ACM/IEEE Design Automation Conference*, 2022. DOI: 10.1145/3489517.3530414.

◇ S. Salehi, **T. Sheaves**, K. I. Gubbi, S. A. Beheshti, D. S. M. P., S. Rafatirad, A. Sasan, T. Mohsenin, and H. Homayoun, "Neuromorphic-enabled security for iot," in *20th IEEE Interregional NEWCAS Conference*, 2022. DOI: 10.1109/NEWCAS52662.2022.9842256.

◇ G. Kolhe, S. Salehi, **T. D. Sheaves**, H. Homayoun, S. Rafatirad, M. P. D. Sai, and A. Sasan, "Securing hardware via dynamic obfuscation utilizing reconfigurable interconnect and logic blocks," in *58th ACM/IEEE Design Automation Conference*, 2021. DOI: 10.1109/DAC18074.2021.9586242.

◇ G. Kolhe, H. M. Kamali, M. Naicker, **T. D. Sheaves**, H. Mahmoodi, P. S. Manoj, H. Homayoun, S. Rafatirad, and A. Sasan, "Security and complexity analysis of lut-based obfuscation: From blueprint to reality," in *IEEE/ACM International Conference on Computer-Aided Design*, 2019. DOI: 10.1109/ICCAD45719.2019.8942100.

◇ A. Attaran, **T. Sheaves**, P. Mugula, and H. Mahmoodi, "Static design of spin transfer torques magnetic look up tables for asic designs," in *Proceedings of the 2018 Great Lakes Symposium on VLSI*, 2018, pp. 507–510. DOI: 10.1145/3194554.3194651.